

intro

- Critical minerals underpin clean energy, electronics, defense, and heavy industry such as steelmaking, fertilizers and chemical production. Because of this, demand for critical minerals is projected to at least triple by 2040. Indium for touchscreens, tantalum for capacitors, rare earths for displays and speakers, lithium and cobalt for batteries, and gallium and silicon for chips.
- china's role in critical mineral refining
 - The International Energy Agency reports that China refines about 90 percent of REEs and is the leading refiner for 19 of 20 key energy-related minerals, with an average global market share of around 70 percent.
 - What is striking is that for many high-demand minerals such as lithium, nickel, and cobalt, China's share of raw extraction is only 10 to 30 percent, yet its share of refining and processing reaches 60 to 70 percent.
 - It also secured foreign mining assets in developing countries through the Belt and Road Initiative. Chinese-backed entities have invested nearly US\$57 billion in mineral projects across 19 countries.²³ This ensures raw material flows into China's refining industry, reinforcing its role as the central hub of global mineral processing.
- where do minerals come from?
 - Australia is also a leading mining nation. In 2023, it supplied nearly half of global lithium and ranked among the top five producers of cobalt, manganese, and REEs.
 - Ukraine holds Europe's largest reserves of lithium and titanium, as well as significant uranium, graphite, manganese, and REEs.³⁶
 - South Africa is central to global supplies of manganese, chromite, vanadium, and platinum-group metals. It holds 80 percent of known manganese reserves and more than 70 percent of chromite. It is also a leading producer of vanadium and platinum.

mobile phone resources (2016)

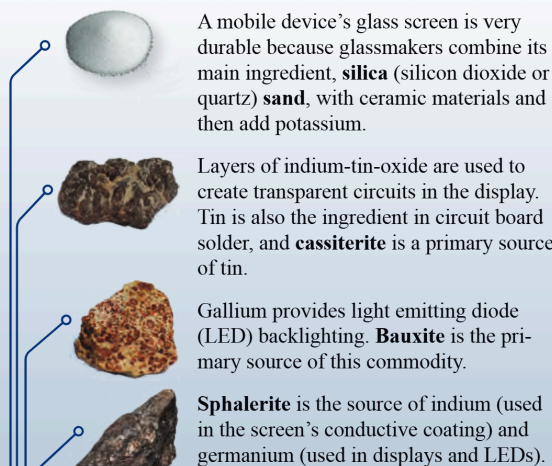
- look:

3 Li lithium 6.94	14 Si silicon 28.08	19 K potassium 39.09	20 C carbon 12.01	29 Cu copper 63.54	31 Ga gallium 69.72	32 Ge germanium 72.63	33 As arsenic 74.92	47 Ag silver 107.86	49 In indium 114.81	50 Sn tin 118.71	73 Ta tantalum 180.94	74 W tungsten 183.84	78 Pt platinum 195.08
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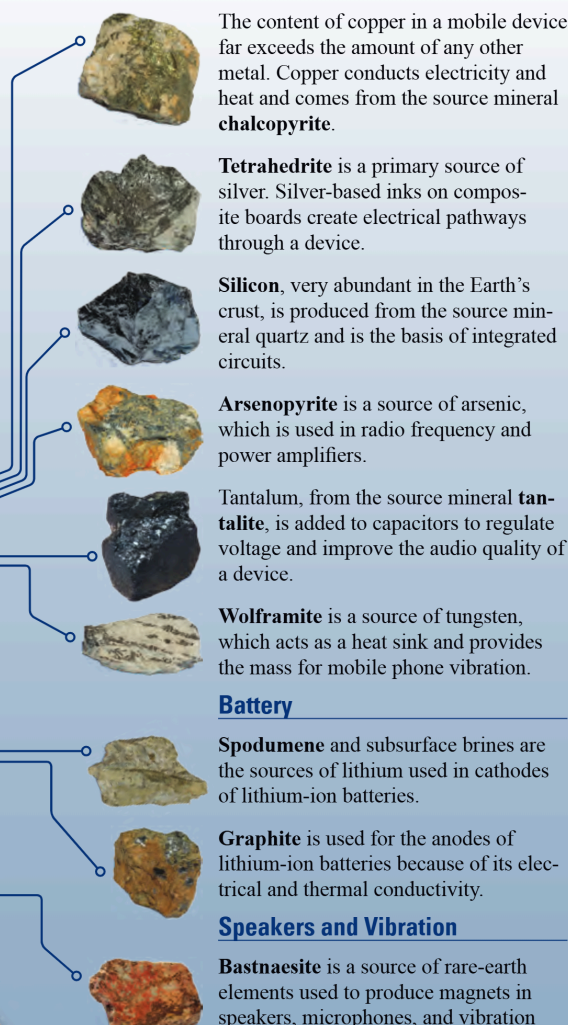
A World of Minerals in Your Mobile Device

Mobile phones and other high-technology communications devices could not exist without mineral commodities. More than one-half of all components in a mobile device—including its electronics, display, battery, speakers, and more—are made from mined and semi-processed materials (mineral commodities). Some mineral commodities can be recovered as byproducts during the production and processing of other commodities. As an example, bauxite is mined for its aluminum content, but gallium is recovered during the aluminum production process. The images below show the **ore minerals** (sources) of some mineral commodities that are used to make components of a mobile device. On the reverse side, the map and table depict the major source countries producing some of these mineral commodities along with how these commodities are used in mobile devices. For more information on minerals, visit <http://minerals.usgs.gov>.

Display



Electronics and Circuitry

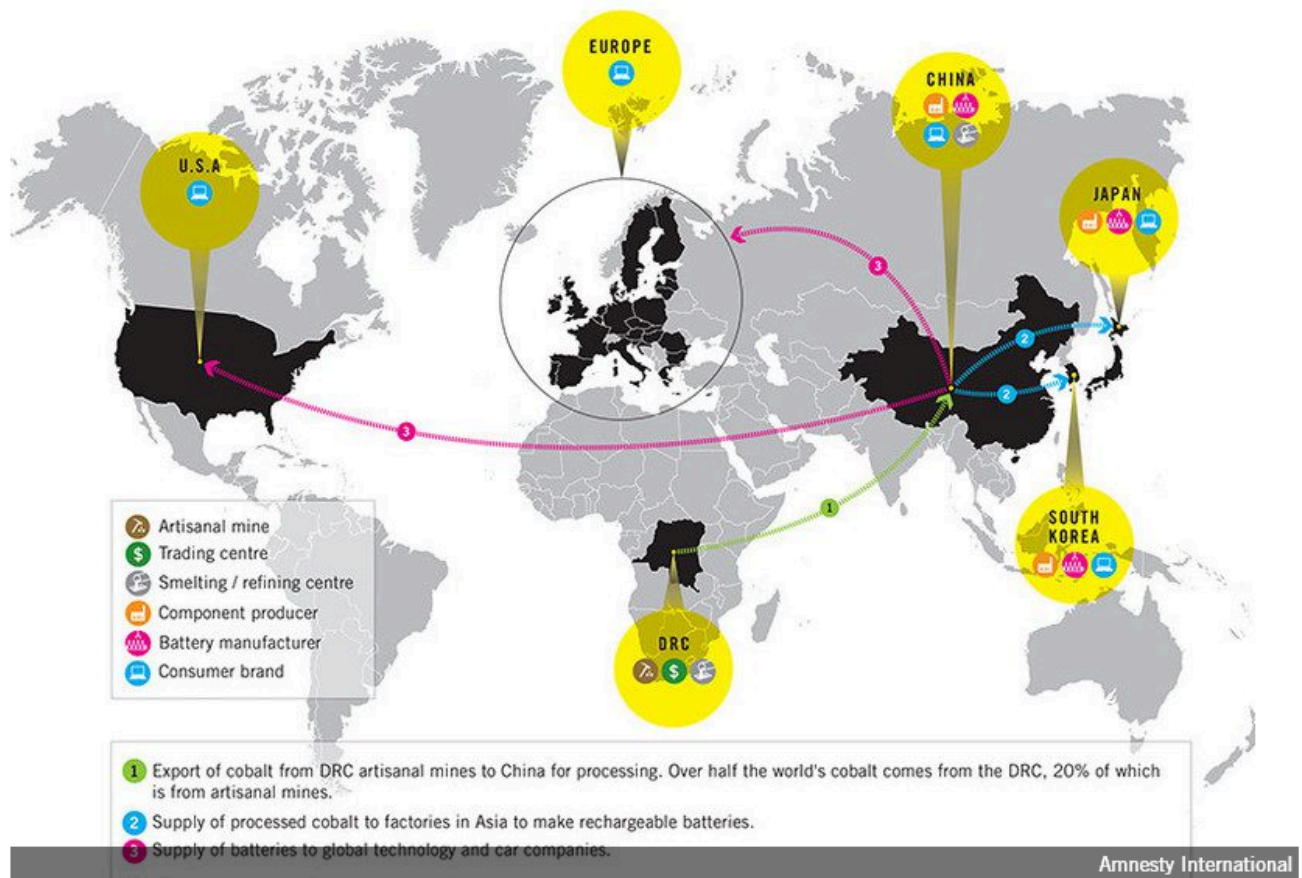


Element	Ores (minerals) occurring in the DRC	Selected uses of element
Copper (Cu)	A variety of Copper sulfide minerals	Electrical wire and cable; Electric motors; Integrated circuits
Cobalt (Co)	Heterogenite, a mixed oxide and Hydroxide of Cobalt (CoOOH)	Alloys for gas turbines and jet engines; Batteries in cell phones and Electric vehicles
Niobium (Nb)	Columbite (also called niobite), an Iron, Manganese, Niobium oxide mineral [(Fe,Mn)Nb ₂ O ₆]	Alloys in jet and rocket engines; Superconducting magnets in magnetic resonance imaging equipment
Tantalum (Ta)	Tantalite, an Iron, Manganese, Tantalum oxide mineral [(Fe,Mn)Ta ₂ O ₆] Columbite-tantalite ores are typically referred to as coltan (see https://friendsofthecongo.org/coltan/)	Capacitors in cell phones and personal computers; Alloys in jet engines
Tin (Sn)	Cassiterite, a tin oxide mineral SnO ₂	Solders for electronic circuits; Alloys with zirconium for cladding of nuclear fuel rods
Tungsten (W)	Wolframite an iron, manganese tungstate mineral (Fe,Mn)WO ₄	Tungsten carbide for cutting tools; Alloys for use in armor- piercing ammunition
Gold (Au)	Native metal, Au	Monetary use; Jewelry; Electrical contacts in personal computers

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cobalt

- byproduct of mining nickel and copper
- DRC - largest producer of cobalt in the world, 60% of world's supply
- 50% of the world's reserve of Cobalt
- 41% of cobalt demand driven by Batteries
- China - leading supplier of cobalt to the US but China gets its cobalt from Congo-Kinshasa
- blood batteries - [amnesty intl investigation coltan 2016](#)
- key mineral for green energy revolution - contributes to manufacture of more eco-friendly products



How cobalt moves from mine to the global market

coltan

- coltan: columbite-tantalite, black-tar like mineral
- DRC - 64% world's reserves of coltan
- when refined, coltan becomes resistant powder that can hold a high electric charge
- vital element in creating devices that store energy / capacitors: mobile phones, computer chips, game consoles
- coltan mined by rebels within DRC and neighbors to DRC (Rwanda, Uganda, Burundi) sold to multi-national corporations, forming "coltan exploitation chain"

3TG: Tin, Tantalum, Tungsten, and Gold

- Tin (cassiterite), tantalum (coltan), tungsten (wolframite), and gold- commonly referred to as the "3TGs"- are the primary conflict minerals that drive continuous civil war in North and South Kivu.
- Together, these resources are necessary for many global industries and products, including smartphones and laptops, energy storage batteries, electric vehicles, artificial intelligence chips, and aerospace and military technologies.
- dozens of militia groups sought to control the lucrative 3TG (tin, tungsten, tantalum, gold) mineral territories in the eastern Congo. The 3TG minerals are vital components to consumer electronic devices and microprocessors.

DRC

- Despite these natural resources, the World Bank estimates that the DRC is among the five poorest nations in the world.³ In 2025, an estimated 74% of Congolese people lived on less than \$2.15 per day.
- On 27 June 2025, the DRC and Rwanda signed a peace treaty, the Washington Accord, to end the conflict in the eastern DRC. The US is meant to provide security guarantees in the eastern Congo in exchange for access to vital mineral territories. Despite the signing, violence and insecurity in the area continues
- MINING
 - Cobalt and copper, key components of green energy technology, are not always labeled as “conflict minerals,” but they play similar divisive roles in the DRC’s south-eastern “Copper Belt.”
 - Fifteen of the 19 major copper-cobalt mining concessions in the DRC are majority-owned by Chinese mining companies

portraits

congo

"I worked in the mines because my parents couldn't afford to pay for food and clothes for me. Papa is unemployed, and mama sells charcoal."

Arthur 13, who worked in a mine from age 9 to 11 years old.

They said that they worked for up to 12 hours a day in the mines, carrying heavy loads, to earn between one and two dollars a day. Even those children who went to school worked 10 – 12 hours during the weekend and school holidays, and in the time before and after school. The children who were not attending school worked in the mines all year around. Paul, aged 14, started mining at the age of 12 and worked in tunnels underground. He told researchers he would often “spend 24 hours down in the tunnels. I arrived in the morning and would leave the following morning.”

mineral pipeline

- Apple: revenue: \$416.2 billion USD (2025), iphone - \$209.6 billion
 - SUPPLIERS
 - Baotou INST Magnetic New Materials
 - annual revenue: CNY 1.44 billion
 - (for magnets, beginning in 2027) MP Materials, Fort Worth, Texas
 - annual revenue: 224.4 million USD
 - vertically-integrated domestic supplier, sourcing minerals from California and Louisiana that are refined local to mines
- Samsung Electronics \$245 billion, smartphones + tablets \$75 billion
 - SUPPLIERS

- claims to source responsible minerals, turn to domestic sourcing and use RMAP-certified smelters
- cobalt -- Glencore and Huayou (the Dark Side of Samsung's Value Chain: the Human Costs of Cobalt Mining "BLOOD, SWEAT AND COBALT")
- Google
 - SUPPLIERS
 - ReElement Technologies, Aqua Metals
 - MP Materials, Fort Worth, Texas
 - 36% recycled minerals, 2030 carbon-free 'moonshot'

ORGANIZATIONS

- **Responsible Minerals Initiative** — responsiblemineralsinitiative.org — industry auditing standards
- PANZI Foundation - Helps Congolese survivors of sexual violence get the healing, justice, community, and future they deserve
- GoFundMe for Goma, DR Congo by Goma Actif (www.gomaactif.org, @gomactif)
- Action Kivu - supports women and youth through trauma healing, vocational programs, and access to education
- Enough Project: aims to stop genocide and human rights abuses in Africa by exposing and disrupting the networks of corruption and violence
- **iFixit** — ifixit.com — right-to-repair as ethical consumption, device longevity